



Lightweight Deep Learning IDS for Resource-Constrained IoT Devices

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Abstract

Internet of Things (IoT) technologies have emerged in the past few years and are used in numerous areas of application such as smart home, healthcare, industrial automation, smart transportation and smart cities. They are generating and processing enormous volumes of information at all times to support automatic processes and decision making. At the same time, with the increasing adoption of IoT networks, there has been a wider attack surface of digital systems that can be targeted by cyber attacks. The limited processing power, memory, storage, and energy resources are typical characteristics of IoT devices. This means that many of the IoT devices are not equipped with a strong security mechanism and are easily prone to security attacks. Distributed Denial of Service (DDoS) attacks, Botnet attacks, Malware injection, Unauthorised access and Data breaches are some of the common cyber attacks for an IoT network. Such attacks can cause service interruptions, steal critical data, and impact the stability of network infrastructures. Hence, helping to provide efficient & effective security solutions for IoT networks is a crucial problem in the Cyber Security field.

To identify and block malicious behaviour and intrusions, Networks require a tool which is known as Intrusion Detection System (IDS). Existing IDS techniques, such as signature and rule-based IDS can be used to detect known attacks. They have a hard time identifying new and advanced cyber threats, though. An automatic feature extraction is a potential solution to effectively recognize the advanced attacks in recent years, and deep learning IDSs have been coming up as an effective solution. In particular, the use of some techniques like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have proven to be effective in detecting abnormal network behaviors. But these deep learning methods require a lot of computational resources, memory and energy. Therefore, it is not possible to use such models on resource-constrained IoT devices. It gives room for the research to design intrusion detection systems with a high detection ability and efficiency in an environment of IoT.

In this study, an efficient deep learning based intrusion detection system (IDS) for low resource IoT environment is proposed. The aim of the research is to lower the complexity of the computation process without sacrificing the efficiency of the intrusion detection by employing feature selection, model simplification, pruning and quantization methods. The proposed system will be trained and tested on popular benchmark datasets like NSL-KDD and UNSW-NB15 to test the performance of the proposed system on various types of attacks. The proposed system will be tested on the metrics of detection and efficiency which are accuracy, precision, recall, F1 score, model size, memory usage and inference time. The lightweight model will also be compared to the traditional deep learning-based IDS models to check the improvements in terms of scalability and efficiency. The expected outcome of this research is to design a scalable and lightweight intrusion detection system that can be easily implemented in real time and deployed on resource-constrained IoT devices to boost the cybersecurity of modern IoT environments.

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1. Chapter 1: Introduction

1.1 Background

Internet of Things (IoT) has brought to life a wide range of applications that feature interconnected devices, such as smart homes, health care, industrial automation and smart cities. Information is collected and sent from IoT devices, helping to automate and make decisions. The advent of an increasing number of IoT (Internet of Things) devices, however, has greatly expanded the attack surface of the networks of today, increasing their susceptibility to cyber attack [9], [20].

The IoT devices typically have limited computational resources, memory and energy resources. However, the security issues of many such IoT based systems are inadequately discussed and they are subjected to security attacks. Some of the most common cyber attacks in an IoT-based network are Denial-of-Service (DoS) attacks, Data theft, Malware injection and Unauthorised access [8], [10], [20]. The attacks can be against devices and network infrastructure (NI).

Intrusion Detection System (IDS) can be a crucial element in monitoring threats to the network traffic. Conventional IDS methods, like signature-based and rule-based systems, depend on pre-defined patterns in order to identify attacks. These techniques are useful for finding known attacks, but are not able to deal with novel attacks and new attacks [3], [26].

Recently, Deep Learning based IDS models have got a lot of attention as they can learn complex patterns from a large number of data sets automatically [2]. Different cyber attack detection techniques such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks are found to be useful for detecting various types of cyber attacks [4], [12] and [13]. Unfortunately, such models are expensive, and are not feasible for deployment on the resource constrained IoT devices.

Moreover, cyber threats are constantly changing, making cyber security of IoT environments more challenging [9], [20]. As threats to data privacy and integrity and availability will increase, cyber adversaries are continually finding ways around traditional security measures. Such a cyber-attack on the IoT devices (which are linked with the critical systems and cloud platforms) can have substantial effects, including outages, financial losses, and privacy violations [20], [21]. The growing number of devices that have been connected and also the growing need to have interoperability has created a need for dynamic and smart security mechanisms to detect known and unknown cyber threats in real-time. Intrusion detection is thus a key part of the security policies of IoT [18], [26].

To overcome these challenges, artificial intelligence (AI) and Deep learning techniques are being increasingly employed in cyber security applications. Deep learning models can be trained to automatically extract meaningful features from extensive network traffic data without needing to manually engineer them, thereby avoiding much manual effort [22]. In current network environments, deep learning methods like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks have proven to perform well at detecting intrusions [4], [12], [13]. However, many of the deep learning-based IDS models are computationally heavy, and are mostly designed for cloud-based or high-performance computing architectures, which are not suitable for deployment on low resource devices such as IoT devices [15], [16], [25]. For this reason, it is increasingly important that efficient and

lightweight intrusion detection systems can be deployed that are able to trade off detection performance for computational efficiency in practical IoT systems.

1.2 Problem Statement

An IDS based on deep learning (DL) has high detection accuracy, but it also has a high demand for processing power, memory, and energy consumption. They are therefore difficult to deploy on resource constrained devices in the IoT world [20], [25]. Furthermore, much of the existing literature does not address efficiency and scalability and limits its studies to improving the accuracy of detection, and few studies consider real-time deployment requirements [5], [15], [25]. Thus, innovative intrusion detection models are required that have high detection accuracy and low computation requirement in IoT environment.

In addition, a lot of the current deep learning-based IDS models are designed and tested in cloud computing or high performance computing systems, where computing resources are abundant. These environments enable complex models to be built for deep learning but do not truly represent real-world IoT deployment conditions [15], [20]. The computational power of IDS models is typically required to be extensive, which is difficult in IoT devices that typically have restricted processing power, memory, and battery life. Therefore, direct application of traditional deep learning IDS models on IoT devices can result in higher energy usage, system overhead and inference delay [17], [24].

A second difficulty is keeping a balance between performance for intrusion detection and computational efficiency. The high effectiveness of lightweight IDS models could result in low computational requirement and easier deployment, however, the model reduction may cause the model to be less reliable for detection and cause false positive or false negative rates to rise [35]. Thus, an efficient deep learning based IDS which has low computational complexity and low false alarm rate to achieve reliable intrusion detection performance in real time IoT applications is needed.

1.3 Research Objectives

This study aims to:

1. To develop a deep learning-based intrusion detection system for IoT networks
2. To create a lightweight model for low-powered devices
3. To achieve high accuracy with low computational overhead
4. To assess the proposed model's performance on benchmark datasets
5. To benchmark the proposed model with existing deep learning-based IDS models

1.4 Research Questions

This study will answer questions such as:

1. What are the techniques to design efficient deep learning-based IDS for resource-limited IoT devices?
2. How can the computational complexity and model size of IDS be minimised while preserving high detection accuracy?
3. How can we balance the model's efficiency (e.g., storage, inference speed) and detection accuracy (e.g., accuracy, precision, recall)?
4. What is the performance of the proposed lightweight IDS model versus that of conventional deep learning IDS models?

2. Chapter 2: Literature Review

2.1 Overview of Intrusion Detection Systems

An Intrusion Detection System (IDS) is an integral part of network security, as it is because it tracks and analyzes network activity to detect any malicious activities or cyberattacks that may occur. IDSs can be broadly categorized as signature-based IDS (SID) and anomaly-based IDS (AID) [27]. There are two types of IDS: Signature-based IDS and Anomaly-based IDS. Signature-based IDS can recognize known attacks by comparing network traffic with a set of predefined attack signatures or patterns, while Anomaly-based IDS can detect abnormal behavior that deviates from normal network traffic [3], [19], [26].

Signature based IDS works well with known attacks and is quite accurate in detecting them. But they have a problem when it comes to detecting new or evolving attacks that don't have a signature. Unlike signature-based IDS, anomaly-based IDS is able to recognize network activity that is abnormal because it learns the patterns of normal network activity and recognizes when these patterns are violated. Although this is beneficial for anomaly-based IDS, however, it is prone to producing many false positives and can lessen its detection ability and operational efficiency in practical situations [19], [26].

Internet of Things (IoT) networks have introduced a significant increase in the scale, variety and complexity of network traffic. Ransomware, Distributed Denial-of-Service (DDoS), and Advanced Persistent Threats (APTs) are all common modern cyber threats that continue to grow more sophisticated—such approaches to IDS are not as effective in a dynamic network environment [9], [20]. For these challenges, IDS systems must achieve high detection accuracy, real-time processing ability, scalability, and low false alarms.

The traditional IDS approach has been proved to be insufficient to detect intrusions, and researchers have turned to machine learning and deep learning methods for intrusion detection instead. Machine learning and deep learning models can learn patterns and behavioral characteristics from network traffic data, without requiring them to be completely programmed by writing rules or signatures, in contrast to rule-based systems [4], [13], [22]. More specifically, deep learning algorithms have been shown to excel at automatically learning complex traffic patterns from massive volumes of data, enhancing the ability of intrusion detection systems to do a better job in the current network environment.

2.2 Deep Learning-Based Intrusion Detection Systems

IDSs are more effective in detecting more complex patterns of data and learning features with deep learning. Some methods like Convolutional Neural Networks (CNN) can be employed to recognize spatial patterns in network traffic and Long Short-Term Memory (LSTM) networks can be used to learn temporal relationships in time series data.

Deep learning based IDS models can be compared to the traditional machine learning based IDS models in some works, which have demonstrated that deep learning is superior to machine learning [12], [13], [28]. CNN-LSTM hybrid models are also suggested to increase the accuracy of IDS. The disadvantage with such models is that they are very expensive to compute and are employed in cloud or high performance computing areas. In the field of Intrusion

Detection System (IDS), the Deep learning methods are used for improving feature learning and increasing detection accuracy [14], [23].

The deep learning technique has been proved to be capable of learning even at a very basic level from the network traffic and has been studied and used extensively in intrusion detection systems (IDSs) [4], [13]. Unlike traditional machine learning techniques, which can involve a lot of feature engineering, deep learning models such as CNN and LSTM can learn complex features from the data.

The capability of deep learning techniques to recognize various kinds of attacks has been demonstrated to be useful recently, especially in detecting advanced or unknown attacks [9]. In the context of temporal attacks, recurrent neural networks (RNNs) and long short term memory networks (LSTMs) have proven to be valuable models of temporal relationships in network traffic, greatly aiding in the detection of time-dependent attacks [4], [12]. However, these models need lots of computational resources like memory and CPU which is difficult to have at hand, especially in resource-constrained environments like IoT networks [16].

Besides, deep learning-based IDS models have been shown to be extremely versatile in detecting known and unknown cyber threats in different network scenarios [22], [29]. As they digest huge amounts of network traffic data and learn from it, they can more easily detect complex and dynamic attack patterns, which are typically difficult to detect. However, as the complexity of the deep learning models continues to grow, the training time of the models, computational costs and energy consumption are growing as well [15], [17]. In the IoT world, with limited processing power and the need for real-time processing for optimal performance, these limitations are more apparent. In recent intrusion detection systems, this has necessitated the need to balance accuracy of detection and computation efficiency in deep learning models, which is a key focus for research.

2.3 Intrusion Detection in IoT Environments

The resource constraints of the IoT devices makes intrusion detection in IoT environment more difficult. IoT devices are small, have limited memory, processing power and power. This is not suitable for deployment of deep learning IDS models on the above devices.

Distributed Denial of Service (DDoS), Botnets and Data Injection attacks are particularly threatening to the IoT networks [8]. So lightweight IDS techniques are becoming increasingly popular that can run within the constrained resources of IoT device and at the same time can be deemed as secure.

The challenges for security in the IoT environments are due to their scale and heterogeneity [20], [21]. The computational power of IoT devices is limited and there aren't robust security measures in place. This puts them at risk of various cyber attacks such as Distributed Denial of Service (DDoS), botnets and data injection. Also, because of the proliferation of IoT, it is hard to apply traditional security solutions. This points to the necessity of having lightweight and efficient Intrusion Detection System (IDS) at the Network Edge or deployed on an IoT device [15], [17], [25].

In addition, scalability and real time processing may be an issue with IDS if these IoT devices are interconnected as they can produce a lot of data [21]. IoT networks are comprised of a

variety of devices with different computational capabilities, communication technologies and operating systems in contrast to the traditional computing environments. It will be more difficult to manage and secure IoT networks. As the number of devices in the network grows, so do the opportunities for cyber-attackers to conduct security threats on IoT systems, indicating the need for effective and flexible security solutions.

Moreover, many IoT devices are used in critical application domains like healthcare, smart transportation systems, smart cities and industrial control systems, where compromising security of such devices could result in disastrous consequences [20]. Successful attacks on these systems can result in service disruption, privacy breach, financial losses and even loss of life. So, IDS should be fast (time and computational) as well as high detection rate, in IoT applications. To overcome the aforementioned problems, an efficient lightweight deep learning IDS, which can provide security for IoT environments and operate under the limited resource of IoT devices, should be developed.

2.4 Lightweight Deep Learning Models

In recent years, the focus has been on lightweight models to minimize computational complexity and overcome the drawbacks of conventional deep learning models. In recent years, the lightweight IDS in IoT networks has been examined, which is aimed at decreasing the computational requirements without compromising IDS detection accuracy [5], [11]. The goal of lightweight models is to ensure that the model size and inference time is small while maintaining the performance of the detector at a reasonable level. In limited resource systems, it would be useful to have such techniques for an IoT application. However, there is a compromise between the speed and accuracy.

In recent years, some efforts have been made to enable the development of models of deep learning in resource-limited settings with efficiency and lightness. Model pruning, knowledge distillation and quantisation are some of the techniques used to reduce the size and complexity of the deep learning models [31], [35]. These are good strategies but there's a need to effectively balance an efficient model with the accuracy of the object detection. Often, lightweight models trade-off their accuracy for improving the speed and memory efficiency. Hence, there is a need to seek the optimum balance of the two in future research.

Moreover, lightweight deep learning models are increasingly being used in edge computing and IoT settings with the need for real-time processing and low latency [33]. For example, in lightweight IDS it could be deployed at the edge or IoT devices without having a server to process cases, or, a cloud-based IDS running on a server processing cases [15], [17], [24]. This reduces network traffic, network latency and enhances privacy by not sending data to a remote server. Thus, the lightweight models of IDS seem to be a more appropriate choice for the distributed IoT systems of today.

In this context, the recent advancement in the optimization of the model has also made the use of deep learning models possible in light weight devices more feasible [29]. To keep the computational cost down and maintain the detection accuracy, several methods have been proven to be effective, including feature learning, parameter compression and network pruning [31], [35].

However, selecting the most appropriate optimization method is not an easy task as the reduction of the model size may also impact on the accuracy and reliability of detection. So, research is still ongoing in the quest for lightweight deep learning models that offer high accuracy and low memory and computational requirements for IoT-based intrusion detection.

Recently, lightweight architectures like MobileNet and TinyML-based models have come to the forefront for edge AI and IoT applications due to their low computational demands and efficiency in their inference speeds [37], [40]. These architectures demonstrate the potential for deploying deep learning models on low-resource hardware platforms while maintaining acceptable detection performance [38], [39]. Such lightweight architectures can serve as design inspiration for efficient intrusion detection systems for the actual deployment of the IoT.

2.5 Limitations of Existing Work

Proposals for deep-learning based intrusion detection systems (IDSs) have been made, but there are still some constraints. Most of the research is related to enhancing the accuracy of the detection without addressing deployment concerns. Computationally complex models are hard to apply to the IoT devices [5], [15]. Furthermore, there are not many studies that have addressed the real-time ID problem in IoT. The current lightweight models are not accurate enough, thus, there is a need for more accurate models to provide efficient and accurate model [27], [28].

Lastly, existing Intrusion Detection System (IDS) is not very good at adapting to other and new IoT networks. Many models are in use and have been tested on relatively small data sets and relatively simple conditions of laboratory tests, where they may or may not resemble the conditions of networks [11], [26] in the real world. The complexity of the network traffics is getting more diversified with the evolution of IoT networks and the complex networks, IDS solutions should be capable of coping with multiple kinds of attacks and be effective. However, many of the solutions proposed are not general enough to deal with the various network environments and attack scenarios, limiting their applicability.

Besides, most of the Deep Learning based IDSs are developed to be highly cloud-based for data processing and storage, which is very important [17] and [21]. Although the cloud offers enough resources for the execution of complex models, it raises other issues such as delays, network bandwidth and privacy issues related to the transfer of sensitive network data. In real time systems of IoT, any delay in the attack detection process can have an effect on the security and reliability [24]. Thus, lightweight IDS solutions are required which can work effectively at the resource constrained IoT environment to detect threats in real-time without having to rely on a centralized control unit.

2.6 Research Gap

While substantial advances have been achieved in the application of deep learning for intrusion detection systems (IDS), several existing methods are yet to address key challenges in IoT networks. Most of the current IDS models focus on achieving very high detection accuracy and employ relatively high computational architectures like deep neural networks (DNNs), convolutional neural networks (CNNs), and hybrid deep learning models [4], [7], [12], [13].

These methods have good detection accuracy in cloud or high performance computing environments, but they often need high computational resources, memory, and energy consumption, which means they are not suitable for resource-limited IoT devices [8], [25].

There have been a number of studies on lightweight IDS models and optimisation methods such as pruning, quantisation, and compressed CNN architectures to facilitate deployment in IoT systems. There are currently a number of lightweight IDS solutions that aim to enhance detection performance but do not consider the deployment aspects including inference time, memory consumption, real-time processing and practical deployment of the IDS on edge devices [5], [6]. Furthermore, several studies assess lightweight IDS performance under simulated or cloud-based scenarios, instead of testing it on real low resource IoT platforms like Raspberry Pi devices [24], [33].

Moreover, the previous lightweight IDS research seldom mentions the performance and efficiency of the security system at the same time, but rather considers security performance and computational efficiency independently. However, lightweight models tend to have a lower detection rate or higher false alarm rates [16]. On the other hand, deep learning models that are very accurate typically consume considerable amounts of compute power and would be impractical for deployment into real-time IoT systems.

Thus, the lightweight deep learning-based IDS that could provide high intrusion detection capabilities and have the potential to be deployed in resource-limited IoT devices via efficient optimisation and deployment-oriented assessment remains in need. In this research, we aim to fill this gap by proposing and testing a lightweight CNN-based IDS by employing optimisation algorithms including structured pruning and quantisation and analysing the feasibility for deployment on Raspberry Pi-based IoT environments.

3. Chapter 3: Methodology

3.1 Overview of Proposed Approach

The goal of this research is to design a lightweight intrusion detection system (IDS) for IoT devices which is resource constrained. The proposed system aims to decrease complexity in the computation process with the high intrusion detection accuracy. The methodology is divided into four key steps: data preparation, model development, lightweight optimization and evaluation. The proposed model will be optimized for deployment on low-resource devices like the Raspberry Pi, which will allow it to run efficiently with limited processing power and memory.

The main goal of the proposed approach is to create an efficient and practical IDS applicable to real-time IoT applications. The study will focus on assessing the efficacy of deep learning technique to enhance IDS performance, decrease the complexity of the model, memory consumption, and inference time of the system [18]. Redundant computational processes will be avoided through lightweight optimisation methods for better deployment possibilities on low resource IoT devices.

Moreover, the proposed lightweight IDS will be tested against the conventional deep learning-based IDS models and the effect of the optimisation techniques on the detection performance and computational efficiency will be checked. The proposed solution will be scalable and flexible for adapting to various network traffic patterns and attack scenarios that are typical for IoT environments [17], [21].

The methodology offers a structured approach for designing, implementing and evaluating the proposed intrusion detection system. The data preprocessing stage will guarantee high quality input data and the model development and optimisation stage will concentrate on building a lightweight and efficient IDS model. Lastly, the evaluation phase will evaluate the detection performance and deployment efficiency in terms of performance and resource based metrics. The idea of this approach is to make the proposed system practical, scalable and appropriate for real-time IoT applications.

3.2 Data Collection and Preprocessing

The proposed model will be tested on popular benchmark datasets like NSL-KDD, and UNSW-NB15 [1], [2]. These datasets are labelled network traffic for normal and malicious traffic. NSL-KDD is chosen because it is structured, eliminates duplicate data from previous data sets and has been widely used in intrusion detection research. It will be used as a reference dataset to test the proposed model for the first time. UNSW-NB15 is added to NSL-KDD because of the more recent and realistic attack scenarios and modern network traffic patterns in it [2], [32].

The preprocessing stage will include:

- Handling missing or invalid data entries
- Scaling features to a common range
- Converting categorical data into numerical format
- Feature selection to remove irrelevant and redundant features

These preprocessing steps help improve model efficiency and reduce computational overhead.

Data exploration will also be performed to understand the distribution of normal and malicious traffic within the selected datasets as well as other characteristics. This process will aid in uncovering class imbalance problems which can impact on model training and intrusion detection [11], [32]. Data balancing and sampling techniques can be used to increase the reliability of a model and its accuracy of detection [32].

The reduction of computational complexity in the context of resource limited IoT environments, and the improvement of model's efficiency, is an important step, and it is known as feature selection [16], [30], [32]. In this research, Information Gain-based feature selection method will be applied to select useful network traffic features and minimize unnecessary data processing. The less number of features you provide as inputs the lower the consumption of memory and the faster the inference will be, which is significant for deployment on low resource devices like the Raspberry Pi [36].

Furthermore, the proposed deep learning IDS model can be enhanced by preprocessing methods like data standardisation, balancing, and filtering, to make the system robust and generalisable. Additionally, efficient IDS can be provided in resource-limited IoT settings through proper preprocessing and feature engineering, which can also minimize the time required for training [40].

3.3 Model Development

In this study, a lightweight deep learning-based intrusion detection system (IDS) in IoT environment will be designed. The main architecture that the proposed IDS will use is a Convolutional Neural Network (CNN), as it is efficient in computation and has good feature extraction ability for structured network traffic data. CNN models are also relatively energy efficient and can be used on resource-limited IoT devices like Raspberry Pi.

The first architecture (CNN) would be convolutional layers followed by fully connected layers and max pooling layers. To reduce the burden of memory requirement and computational efficiency, lightweight configurations will be studied to keep the number of trainable parameters to an acceptable level, while maintaining suitability for intrusion detection performance.

CNN is chosen as the main model due to its computational efficiency and its ability of extracting features for structured network traffic data well [37]. For comparison, LSTM and hybrid CNN-LSTM architectures can be considered. LSTM networks have proven useful for traffic pattern analysis in sequences; automatic feature extraction combined with temporal analysis power is provided by hybrid CNN-LSTM networks [10]. The baseline models (CNN, LSTM, CNN-LSTM) will be compared with the proposed lightweight CNN model using the accuracy, precision, recall, F1-Score, inference time and computational cost metrics.

The model architecture will be tailored for low resource hardware infrastructure of IoT. The number of hidden layers, activation functions, learning rate, batch size, and trainable parameters will be fine-tuned to enhance the detection accuracy while minimizing the computation. The goal of this optimisation process is to create a lightweight and efficient IDS

which can be deployed to real-time environments in an IoT, with limited resources [12], [33], [34].

3.4 Lightweight Optimization Techniques

To adapt the proposed IDS model for resource-constrained IoT devices, several lightweight optimisation techniques will be applied:

- Feature Selection: Reducing unnecessary input features to lower computational overhead
- Model Simplification: Reducing the number of layers and trainable parameters
- Structured Pruning: Removing less significant network connections to reduce model complexity
- Post-training 8-bit Quantisation: Reducing weight precision from 32-bit to 8-bit to decrease memory usage and inference time [16]

The optimisation techniques enable to reduce memory usage, computation complexity and inference time with acceptable intrusion detection performance [5], [16], [35].

Lightweight optimisation has been especially essential to deploy deep learning across limited processing power, memory and energy devices like IoT devices. The effect of each optimisation technique on the detection performance and computational efficiency will be tested in this research. The study will examine the accuracy, model size, memory consumption, inference speed trade-off to determine the optimized lightweight IDS configuration for real-time IoT applications.

Given these use cases, and the need for lightweight deployment and inference on edge devices, TensorFlow Lite will be ported to support these devices. The optimisation techniques that are considered lightweight can also contribute to reduction of the energy consumption, for battery-powered IoT devices, which is important too [15], [16].

Another relevant fact is the stability of detection following the compression of the model. Over-reduction of models could have a negative impact on intrusion detection and higher false positive rate. Thus, in this research, performance analysis of the balance between the efficiency of computing and the accuracy of intrusion detection will be conducted to guarantee the IDS performance in the real-time system on resource-limited IoT environments.

The overall goal of the lightweight optimisation phase is to enhance the deployability, scalability and practicality of the proposed deep learning-based IDS without compromising intrusion detection performance.

3.5 Model Training and Evaluation

The proposed model will be trained and evaluated using benchmark datasets such as NSL-KDD and UNSW-NB15 [1], [2]. The model performance will be assessed using the following detection metrics:

- Accuracy
- Precision
- Recall

- F1-score

In addition, efficiency-based metrics will also be evaluated:

- Model size
- Training time
- Inference time
- Memory usage

The proposed lightweight IDS model will also be compared to the conventional deep learning based IDS models to assess the effect of lightweight optimisation techniques on the accuracy of the detection and computational efficiency.

The data sets will be split in a ratio of 80:20 or a similar ratio. Optimisation algorithms like Adam will be used to train the model, and cross validation methods will be employed to enhance the generalisation capability and reliability of the model.

The proposed intrusion detection system will be tested in order to prove its ability to accurately detect the normal and malicious traffic with different attack scenarios and network traffic. True positives, true negatives, false positives and false negatives will also be determined through the use of a confusion matrix analysis [26], [32].

Performance monitoring will be used for training to avoid over fitting, under fitting and to analyse model convergence and loss reduction. To enhance model stability and generalisation capability, techniques like dropout and regularisation and early stopping can be employed [4], [13]. The hyperparameters such as learning rate, batch size and number of hidden layers will also be tuned to determine the best configuration of the model for a IoT intrusion detection [23].

Further, the lightweight optimised model will be evaluated by deployment-oriented metrics like model size, memory usage and inference speed to check its suitability for deployment in the real-time deployment of an IoT device like Raspberry Pi [15], [16], [25].

In summary, the training and evaluation phase is designed to validate the proposed lightweight IDS to provide high intrusion detection performance with low computation complexity and low requirement for deployment in real-world IoT environments.

3.6 Tools and Environment

The proposed intrusion detection system will be implemented using the following tools and technologies:

- Programming Language: Python
- Deep Learning Framework: TensorFlow/Keras
- Deployment Framework: TensorFlow Lite
- Development Platform: Jupyter Notebook and Google Colab

The choice of Python as the primary programming language stems from the fact that it is widely used to develop machine learning and deep learning applications. The development, training,

and assessment of the model will be done with TensorFlow/Keras and deployment and inference on the Raspberry Pi devices will be supported with TensorFlow Lite.

Jupyter Notebook or Google Colab will offer interactive workspaces for developing and experimenting with models and analyzing their performance. Data preprocessing, feature selection, statistical analysis and result visualisation will be provided using extra libraries like NumPy, Pandas, Scikit-learn and Matplotlib.

These tools offer a flexible and efficient space for developing and testing lightweight IoT-based IDSs based on deep learning.

3.7 Proposed System Architecture

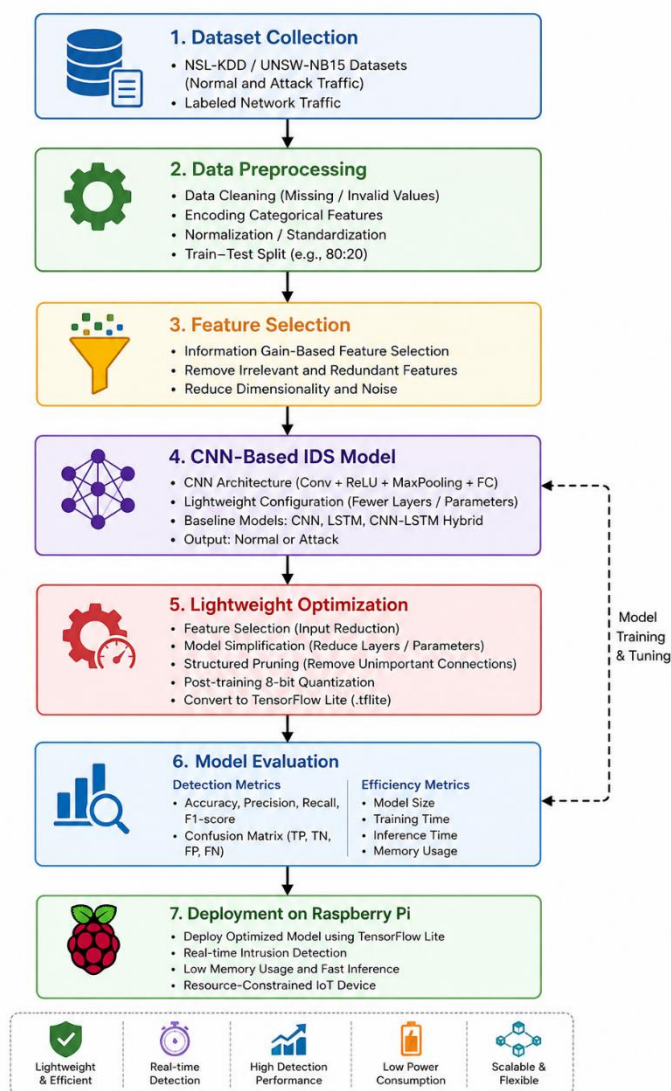


Figure 1: Proposed Lightweight Intrusion Detection System Architecture

The proposed lightweight deep learning based intrusion detection system (IDS) architecture for resource limited IoT devices is shown in Fig. 1. The process starts with the selection of

benchmark datasets, e.g., NSL-KDD and UNSW-NB15, that include labelled network traffic data both for normal and malicious activities [1], [2]. Such datasets offer appropriate training and test sets for the evaluation of intrusion detection performance for various attack scenarios.

Data preprocessing is the next step, which includes data cleaning, standardization, converting categorical data to numerical data, and scaling features to a common scale. Feature selection methods are then used to eliminate irrelevant and redundant features and discover the most significant network traffic features. This can help lower the computational complexity and enhance the efficiency of the model, especially in an IoT environment.

The preprocessed data is then used to train a Convolutional Neural Network (CNN) to identify network traffic behaviour patterns and identify malicious activities [4], [12], [13]. Lightweight optimisation techniques like structured pruning, post-training quantisation and model simplification are used to reduce computation complexity, model size and memory consumption to ensure that the intrusion detection performs well on resource constrained IoT devices [36].

The optimised model will then be saved in TensorFlow Lite format to deploy and infer on Raspberry Pi devices. Both detection and efficiency will be used to evaluate the proposed IDS. The metrics for evaluating the detection will include accuracy, precision, recall, and F1-score, and the metrics for evaluating the efficiency will include inference time, memory usage, and size of the model to determine their suitability for real-time IoT applications.

Moreover, the proposed architecture can be scalable and flexible in various IoT network environments and attacks. Because of the modular structure, the stages of preprocessing, model development, optimization and evaluation can be modified or enhanced according to the future research needs and technological advances [17], [21].

The proposed system architecture offers a structured and efficient way to build a lightweight and scalable IDS that can be used in real-time IoT cybersecurity applications [15], [16], [35].

4. Expected Outcomes

This research is expected to develop a lightweight deep learning intrusion detection system, which can be used in resource constrained IoT devices. This model is anticipated to decrease the computational and memory complexity, decrease time taken for making inferences and ensure accurate detection. The proposed model is expected to improve computational efficiency and reduce model size while maintaining strong intrusion detection performance. The F1-scores of the proposed model are expected to be better than 85% while the size of the model is expected to be reduced by 30-50% from the conventional deep learning based IDS models [15], [16], [34], [35].

Furthermore, it will also decrease the inference time and enable real-time intrusion detection for low resource devices, e.g., Raspberry Pi. In a previous study, it was proven that lightweight models optimization techniques such as pruning, quantisation and knowledge distillation can effectively reduce the model size and computation without compromising the intrusion detection system performance, while being cost-effective [16], [25], [35].

The proposed research will likely result in an efficient lightweight intrusion detection system with a high capability of detection within a low computational complexity, memory consumption, and inference time. The research will also investigate the ability to manage the performance of intrusion detection vs computational efficiency to ensure the feasibility of the proposed model to be deployed in real time on Raspberry Pi-based IoT environments. Moreover, the research will showcase the feasibility of deployment for low-resource edge devices by using lightweight optimisation methods like structured pruning and quantisation.

The study will also surely be compared with the traditional deep learning based IDS models with the proposed lightweight IDS. The results can provide some insights on the trade-off between model complexity, performance and efficiency [33, 35]. The findings could pave the way for more efficient deep learning models to be designed for resource-constrained devices and could help to make the realization of lightweight security systems in practical IoT applications easier.

Finally, the proposed intrusion detection system can be beneficial for security and trust of the modern IoT network in various IoT applications such as smart home, smart healthcare, industrial automation, and smart city applications. The goal of the research is to identify threats in real-time efficiently on resource constrained devices, and to be able to implement effective cyber security solution, which will be capable of solving some of the emerging cyber threats in modern networked devices.

5. Timeline

The proposed timeline for this research is outlined below. It provides a structured plan for completing the key stages of the project. The timeline will be refined further based on discussion with the supervisor to ensure feasibility and alignment with project requirements.

Phase 1: Literature Review & Problem Understanding

- Review existing IDS and deep learning models
- Study IoT security challenges
- Identify research gap and finalize scope

Phase 2: Dataset Analysis & Preprocessing

- Study NSL-KDD and UNSW-NB15 datasets
- Perform data cleaning and normalization
- Feature selection and preprocessing

Phase 3: Model Development

- Develop baseline model (CNN)
- Train and test initial model
- Analyze baseline performance

Phase 4: Lightweight Optimization

- Apply pruning, simplification, and quantisation
- Reduce model complexity
- Optimize for IoT deployment

Phase 5: Evaluation & Comparison

- Evaluate using accuracy, precision, recall, F1-score
- Measure model size and inference time
- Compare with baseline model

Phase 6: Final Report submission

- Compile results and analysis
- Compile entire thesis report
- Review and finalize submission

Research Activity	Timeline						
	30 May – 14 Jun 2026	15 Jun – 30 Jun 2026	1 Jul – 31 Jul 2026	1 Aug – 31 Aug 2026	1 Sep – 30 Sep 2026	1 Oct – 31 Oct 2026	1 Nov – 30 Nov 2026
1. Literature Review & Problem Understanding							
2. Dataset Analysis & Preprocessing							
3. CNN Model Development							
4. Lightweight Optimization							
5. Evaluation & Comparison							
6. Final Report Submission							
Time Period	30 May 2026 to 30 November 2026						

Figure 2: Proposed Research Timeline

6. Conclusion

The goal of this research is to build a lightweight deep learning-based intrusion detection system for the resource-constrained IoT environment. The study concentrates on more feasible deployment by reducing the computational complexity, amount of memory needed, and inference time that is required to generate intrusion detection performance while keeping it strong. The proposed method is to fuse CNN with low complexity optimization techniques like structured pruning, quantisation, and model simplification to enable efficient deployment in realtime on Raspberry Pi devices.

The research also seeks to assess the detection/Computation efficiency trade-off based on security and deployment metrics. The proposed system will incorporate lightweight optimization and edge deployment factors, making it both scalable and practical for emerging IoT deployments. In general, this study will help develop effective and deployable deep learning cybersecurity solutions for a low-resource IoT platform.

7. References

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